Gyandeep Kumar

Mobile • Web • TV

React Native | TypeScript | iOS & Android | WebRTC & CallKit | Firebase & Cloud | Stripe & POS Payments | Marketplace & Bookings | App Store & Play Releases

SELECTED WORK • CASE STUDY

Boston Payroll — Enterprise Billing, Stripe Payments & Multi-Tenant Platform

Project Overview

Boston Payroll is an enterprise-grade mobile administration and multi-state compliance platform engineered for business owners, CPAs, and HR managers. Built with a high-performance **React Native (iOS & Android) client**, the application manages the complete operations lifecycle—from real-time multi-tenant telemetry and flexible pay run execution to automated **QuickBooks Online accounting sync**, itemized invoicing with **Stripe payment checkout**, and state/federal tax liability reporting.

Key Client Solutions & Technical Impact

STRIPE & BILLING — Instant Stripe Links

Challenge: Bureau operations suffered from manual invoice creation, payment chasing, and disconnected recurring billing across hundreds of client companies.

- ✓ **Automated itemized invoice generator with subscription tier billing ($199/mo Plus, $299/mo Pro) and direct Stripe checkout links.**

TAX COMPLIANCE — 99.9% Audit Accuracy

Challenge: Employers faced severe IRS and state penalty risks from inaccurate withholdings and multi-state tax computations.

- ✓ **High-precision automated calculation engine tracking VI and Federal unemployment liabilities with 1-tap audit-ready exports.**

ACCOUNTING SYNC — 100% Automated Ledger

Challenge: Accountants and CPAs wasted 15+ hours each month manually re-keying payroll batches and tax deductions into accounting software.

- ✓ **1-tap bi-directional QuickBooks Online API integration synchronizing payroll, contractor payouts, and tax liabilities into journal entries.**

WORKFORCE TRACKING — Zero Timesheet Disputes

Challenge: Managers struggled with attendance discrepancies, missed punch cards, and unapproved overtime across distributed workforces.

- ✓ **Real-time clock-in/out tracking with automated missed punch detection, timesheet dispute workflows, and multi-tier leave approvals.**

Core Features & Engineering Highlights

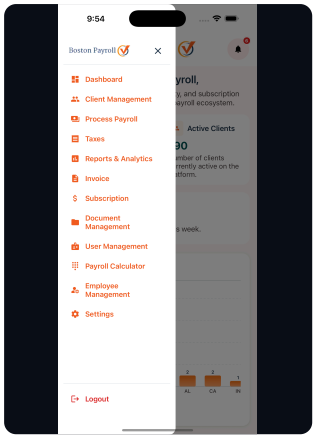
- **Multi-Tenant Dashboard & Client Telemetry:** Executive telemetry tracking total and active client entities, weekly scheduled payroll runs, and state-by-state distributions.
- **Stripe Payment Gateway & Itemized Invoicing:** Transparent billing engine managing recurring subscription plans ($199/mo Plus, $299/mo Pro) with instant Stripe payment links.
- **QuickBooks Online Bi-Directional Accounting Sync:** Direct integration synchronizing employee and contractor payroll entries alongside liabilities into accounting journals.
- **Automated Pay Schedule & Processing Engine:** Flexible configuration engine supporting weekly, bi-weekly, semi-monthly, and monthly pay runs with custom deductions.
- **End-to-End Tax Compliance & 1-Tap Audit Export:** Real-time tax payment records, VI Unemployment, and Federal Unemployment calculations with one-tap audit export.
- **Workforce Attendance & Real-Time Punch Correction:** Live clock-in tracking, automated missed punch detection, timesheet dispute resolution, and overtime tracking.

Application Screenshots



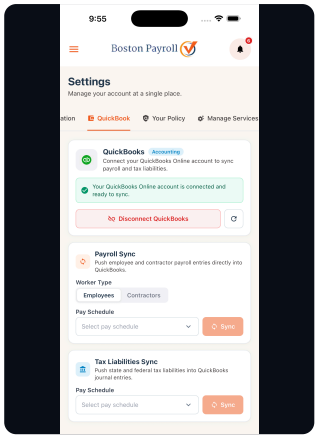
SCREEN 01

Executive Payroll Dashboard



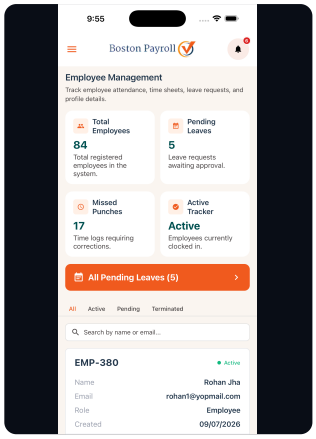
SCREEN 02

Navigation & Module Drawer



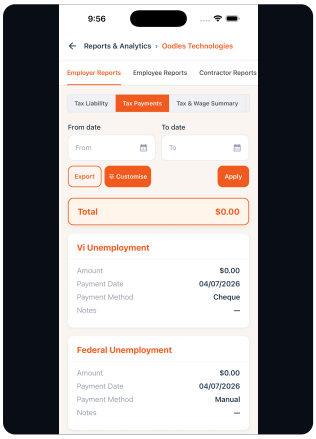
SCREEN 03

QuickBooks Online Sync



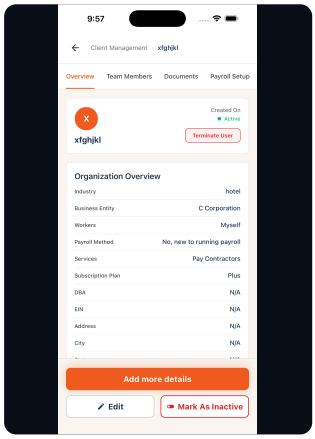
SCREEN 04

Employee Attendance Roster



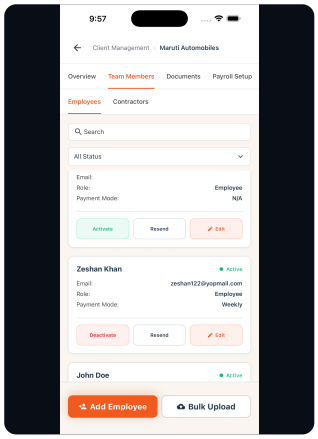
SCREEN 05

Tax Payments & Reports



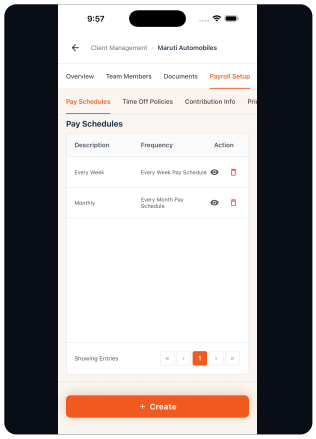
SCREEN 06

Organization Overview



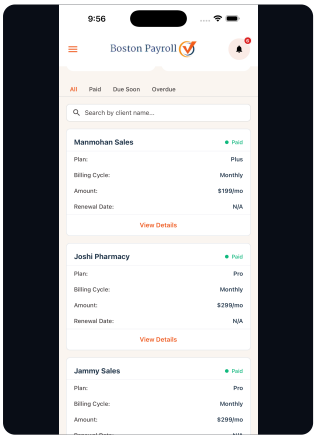
SCREEN 07

Workforce & Contractors



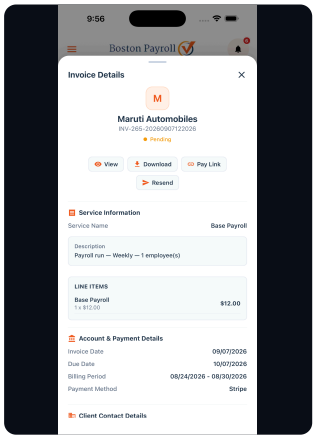
SCREEN 08

Pay Schedules Engine



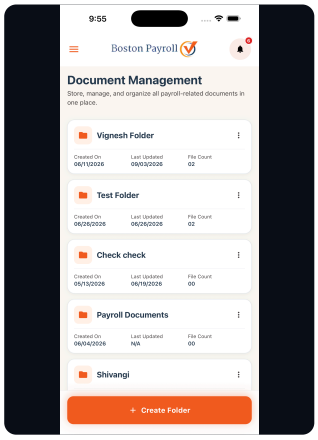
SCREEN 09

Subscriptions & Billing



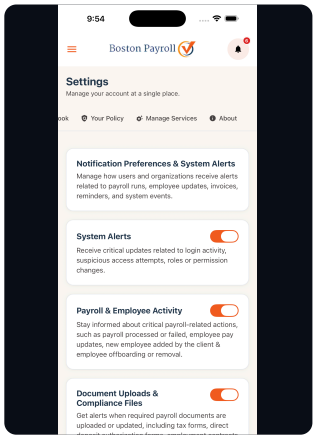
SCREEN 10

Itemized Invoice & Pay Link



SCREEN 11

Compliance Document Vault



SCREEN 12

Audit Alerts & Settings

SELECTED WORK • CASE STUDY

Eventos Arena — Event & Marketplace Mobile Platform

Project Overview

Eventos Arena is an enterprise-grade mobile companion and two-sided event marketplace application engineered for large-scale expositions and conventions. Built with a high-performance **React Native (iOS & Android) mobile client** and robust cloud synchronization, the platform connects event hosts and organizers with attendees through real-time multi-event discovery, interactive vector venue floor maps, instant contactless QR digital passes, direct 1-on-1 B2B session booking, and live agenda management.

Key Client Solutions & Technical Impact

B2B MARKETPLACE — Pre-Booked Sessions

Challenge: Attendees and exhibitors lacked a direct channel to discover partners, schedule 1-on-1 meetings, and manage session bookings.

- ✓ **Engineered B2B matchmaking hub with calendar booking, attendee discovery & privacy controls.**

QUEUE-FREE ACCESS — <0.5s QR Turnstile Scan

Challenge: Event organizers struggled with 45+ minute venue entry queues and manual paper registration choke points.

- ✓ **Single-tap cryptographic QR digital pass on mobile home screen for instantaneous, contactless check-in.**

VENUE NAVIGATION — 180+ Interactive Booths

Challenge: Visitors experienced disorientation finding specific exhibitor booths and stage tracks across multi-pavilion halls.

- ✓ **Interactive vector floor plans with multi-pavilion switching, dynamic booth search, and real-time routing.**

REAL-TIME SYNC — Instant Push Alerts

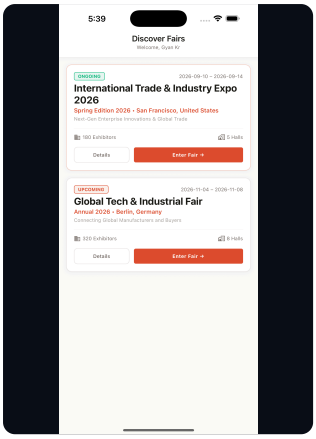
Challenge: Organizers needed to push real-time agenda changes, emergency announcements, and live event updates.

- ✓ **Bi-directional cloud synchronization with push notification infrastructure (APNs/FCM) for live updates.**

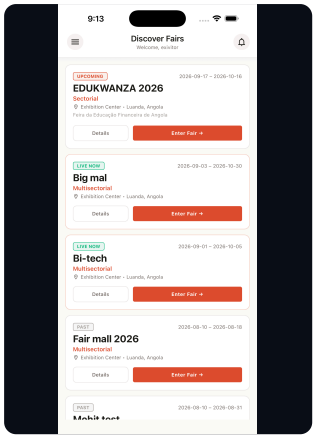
Core Features & Engineering Highlights

- **Interactive Pavilion & Floor Plan Navigator:** High-resolution vector floor maps displaying booth allocations, multi-hall layouts, air-conditioned zones, and emergency routes.
- **Single-Tap Contactless Digital Pass:** Cryptographic QR badge integrated directly into the home screen for queue-free turnstile entry and exhibitor badge scanning.
- **Multi-Event Marketplace & Lifecycle Engine:** Dynamic multi-expo catalog with real-time status indicators (Live Now, Upcoming, Past) for localized and international fairs.
- **Host & Exhibitor Directory with 1-on-1 Meeting Scheduler:** Searchable brand directory categorized by pavilion zones with direct 1-on-1 meeting appointment booking.
- **Multi-Track Session Agendas & Bookmarks:** Multi-day keynote and panel schedule with filtering across Keynotes, Panels, and Networking tracks.
- **Real-Time Cloud Sync & Push Notification Delivery:** Bi-directional synchronization of event rosters, credentials, and instant push notices across devices.

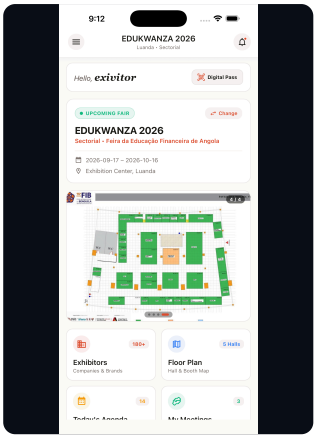
Application Screenshots



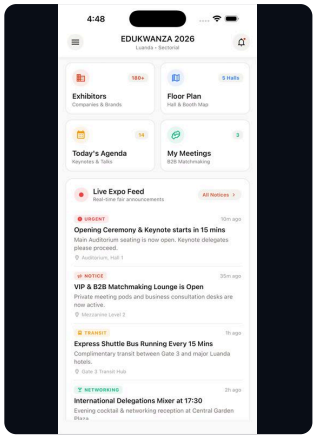
SCREEN 01
Discover Fairs Feed



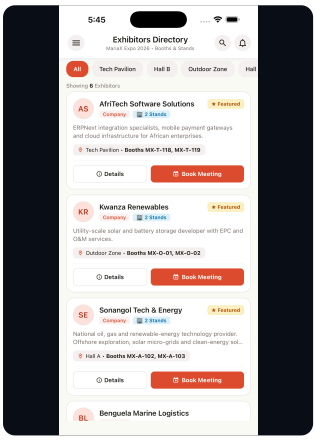
SCREEN 02
Live Status Manager



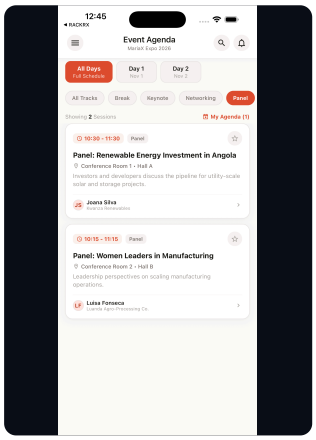
SCREEN 03
Floor Plan & Digital Pass



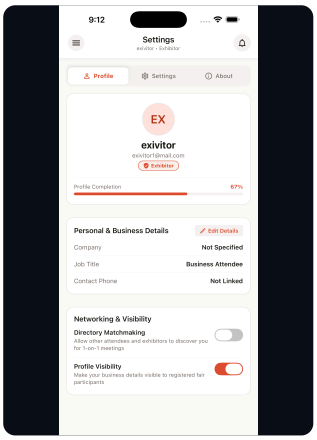
SCREEN 04
Live Expo Broadcast Feed



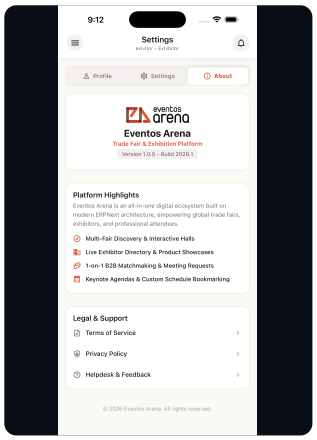
SCREEN 05
Exhibitors & Book Meeting



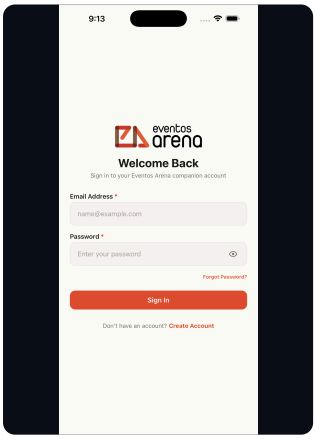
SCREEN 06
Keynote & Panel Agendas



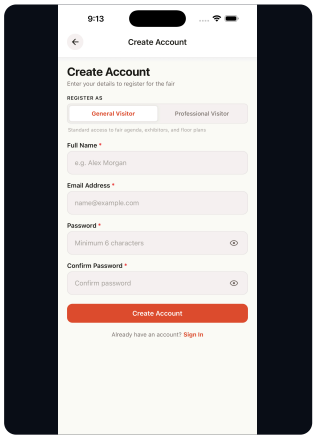
SCREEN 07
Networking Profile



SCREEN 08
ERPNext Architecture



SCREEN 09
Secure Sign In Portal



SCREEN 10
Role-Segmented Onboarding

SELECTED WORK • CASE STUDY

Meds — Healthcare POS, Inventory & Geofenced Attendance Platform

Project Overview

Meds is an enterprise healthcare point-of-sale (POS) and pharmaceutical inventory platform engineered for retail pharmacy chains and medical dispensaries. Built with **React Native (iOS & Android)**, **Redux Toolkit**, and a real-time **WebSockets event pipeline**, the application accelerates counter checkouts through rapid drug catalog search, tracks multi-warehouse batch numbers and expiration dates, automates GPS geofenced staff attendance verification, and streams real-time order dispatch telemetry with Firebase Cloud Messaging (FCM).

Key Client Solutions & Technical Impact

POS VELOCITY — <15s Checkout Time

Challenge: Retail pharmacists experienced long customer queues during peak hours due to slow item lookups and manual pricing calculations.

- ✓ **Engineered barcode-style drug search and prescription cart with batch selection, automated discounts, and instant receipt generation.**

INVENTORY INTEGRITY — Zero Expired Dispensing

Challenge: Pharmacies suffered heavy revenue losses and safety risks from expired inventory and untracked lot quantities.

- ✓ **Batch-level SKU tracking engine with automatic expiration date alerts and low-stock reorder triggers.**

STAFF GEOFENCING — 100% On-Site Verified

Challenge: Pharmacy managers struggled with unverified off-site clock-ins and attendance discrepancies across multi-branch locations.

- ✓ **High-precision device geolocation validating staff presence within physical branch boundaries before recording punches.**

LIVE ORDER PIPELINE — Real-Time Dispatch Sync

Challenge: Disconnected order communication between front-counter POS stations and backoffice inventory fulfillment runners.

- ✓ **Bi-directional WebSockets middleware streaming instant order events with Firebase push notification dispatch.**

Core Features & Engineering Highlights

- **High-Speed Point of Sale (POS) Engine:** Instant medicine search, prescription-backed cart management, customer tagging, tax/discount calculation, and receipt generation.
- **Batch & Expiration Date Stock Tracker:** Granular batch number management, automated expiry warnings, minimum threshold alerts, and inventory transfers.
- **GPS Geofenced Employee Attendance Tracker:** High-precision device geolocation boundary verification, automated shift timecard logging, and overtime tracking.
- **WebSockets-Powered Live Order Dispatch Pipeline:** Real-time event streaming synchronizing order status between counter POS stations and inventory fulfillment runners.
- **Executive Pharmacy Sales & Performance Telemetry:** Real-time gross revenue tracking, OTC vs. prescription breakdown, top-moving medication analytics, and daily profit margins.
- **Customer CRM & Prescription Refill Vault:** Searchable patient directories, historical medication purchase logs, automated refill reminders, and verified phone records.

SELECTED WORK • CASE STUDY

SeetBelt (Careh PA) — Family Safety, WebRTC Video & Emergency SOS Platform

Project Overview

SeetBelt (Careh PA) is a mission-critical family safety, remote caregiving, and emergency response mobile application engineered for real-time protection and rapid distress escalation. Built with **React Native (iOS & Android)**, **WebRTC peer-to-peer audio/video streaming**, native **iOS CallKit**, and **Android lock-screen call managers**, the platform delivers one-touch Emergency SOS dispatch, live GPS geofencing alerts, full-screen alarm takeovers, check-in countdown timers, and parent-to-child safety monitoring.

Key Client Solutions & Technical Impact

EMERGENCY RESPONSE — <1s Distress Broadcast

Challenge: Users in distress or medical emergencies could not navigate complex unlock screens or multiple apps to call for help.

- ✓ **Single-touch SOS button triggering immediate high-priority push broadcasts, GPS coordinate transmission, and emergency contact escalation.**

LOCK-SCREEN OVERRIDE — 100% Alarm Penetration

Challenge: Critical safety check-in alerts and family distress calls were missed when devices were muted, locked, or in Do Not Disturb mode.

- ✓ **Engineered native Swift & Kotlin foreground services overriding system audio and presenting full-screen lock-screen alarms.**

SUB-SECOND VIDEO — Peer-to-Peer Stream

Challenge: Standard cellular calls failed during remote monitoring scenarios requiring instant verified video check-ins and two-way audio.

- ✓ **Integrated AntMedia WebRTC engine with native CallKit (iOS) and custom incoming call UI (Android) for peer-to-peer video sessions.**

AUTOMATED GEOFENCING — Real-Time Safe Zones

Challenge: Parents and caregivers needed proactive alerts when family members arrived at or departed from designated safety zones (home, school, clinic).

- ✓ **Background geolocation engine tracking geofence transitions with low-battery consumption and real-time push notifications.**

Core Features & Engineering Highlights

- **One-Touch Emergency SOS & Distress Dispatch:** Immediate high-priority distress broadcast with live GPS breadcrumb tracking, siren triggers, and emergency contact alerts.
- **Full-Screen Lock-Screen Alarm & Check-In Engine:** Native foreground services taking over locked devices with progressive audio sirens and mandatory biometric check-ins.
- **WebRTC Peer-to-Peer Video & Voice Calling:** Sub-second audio/video communication with native iOS CallKit integration, custom Android call managers, and noise suppression.
- **Real-Time Geofence Safety & Boundary Alerts:** Background location engine monitoring designated safe zones (home, school, hospital) with instant entry/exit push notifications.
- **Parent-to-Child Remote Monitoring Cockpit:** Centralized safety dashboard displaying battery status, live location, connectivity telemetry, and check-in history.
- **Cross-Platform Foreground Services & Wake-Lock:** Native Swift and Kotlin background daemon management ensuring zero missed alerts even when the app is killed or sleeping.